Automata and Computability (Quiz 3)

Course Code: CSE331
Total Marks: 20
Section 17

ID: __________________ **Name:** _______________________ **Class Roll:** _________

Questions

Q1. (9 Marks) Let $\Sigma = \{0, 1\}$. Prove that the following language is not regular:

L = $\{www : w \in \Sigma^\star\}$

Q2. (9 Marks) Let $\Sigma = \{1\}$. Prove that the following language is not regular:

L = $\{w \in \Sigma^\star : w = 1^{n^3+n}$,where n ≥ 0 $\}$

Q3. (2+2+2+1 = 7 Marks) Let $\Sigma = \{0, 1\}$.

$$S \rightarrow P0Q$$
$$P \rightarrow 0P1 \,|\, 1P0 \,|\, 1P1 \,|\, 0P0 \,|\, 0 \,|\, 1 \,|\, \epsilon$$
$$Q \rightarrow 0Q \,|\, 1Q \,|\, \epsilon$$

(a) Give a leftmost derivation for the string '11000'.

(b) Draw the parse tree corresponding to the derivation you gave in (a).

(c) Show that the given grammar is ambiguous by drawing another parse tree (*apart from the one that you have drawn in (b)*) for the given string in (a).

(d) Write one five-letter string that has exactly one parse tree in the given grammar.